

Luigi Gargioni

Nazionalità: Italiana **Data di nascita:** 26/06/1993

Numero di telefono: (+39) 3343404482

Indirizzo e-mail: luigi.gargioni@gmail.com **Skype:** gargio26@hotmail.it

Whatsapp Messenger: +39 3343404482

LinkedIn: https://www.linkedin.com/in/luigigargioni

Abitazione: Via Piave 35, 25023 Gottolengo (Brescia) (Italia)

ESPERIENZA LAVORATIVA

Dottorato di RIcerca
Antares Vision [01/12/2022 – Attuale]

Città: Travagliato, Brescia
Paese: Italia

Telemedicina e Digitalizzazione dei Servizi Sanitari

Ingegnere sviluppo software - Libero professionista
Gargioni Luigi [12/2022 – Attuale]

Città: Brescia
Paese: Italia

• Sviluppo software Full-Stack
• Sito web: https://luigigargioni.github.io/

Scoutman
TieBreakTech [10/2019 – Attuale]

Città: Montichiari (Brescia)
Paese: Italia

Scoutman per la Lega Pallavolo Serie A Femminile presso il PalaGeorge di Montichiari (BS) e il PalaRadi di Cremona.

Attività svolte:
• Raccolta statistiche e collaborazione con gli addetti stampa.

Software Engineer - R&D Developer
Kline s.r.l. [16/11/2019 – 30/11/2022]

Città: Leno (Brescia)
Paese: Italia

Kline sviluppa e produce piattaforme software integrate, soluzioni tecnologiche e servizi per gli intermediari finanziari.

Dipartimento di Ricerca e Sviluppo - Attività svolte:
- Reengineering applicazione esistente con paradigma di programmazione funzionale e tecnologie web: JavaScript, TypeScript, Webpack e React (Redux, Styled Components, Jest, ...).
- Product Owner di due progetti in outsourcing con una società internazionale
- Selezione del personale

Precedentemente - Analista Programmatore:
- Analisi funzionale di procedure in ambito bancario e finanziario.

- Implementazione dei moduli in Delphi, Java e PL/SQL (Oracle DB).
- Attività di supporto ai clienti in relazione agli applicativi sviluppati e distribuiti.

Collaboratore
Punto Pizza [27/02/2011 – 29/12/2021]

Città: Gambara (Brescia)
Paese: Italia

Pizzeria d'asporto e al taglio.

Attività svolte:
- Aiuto pizzaiolo e addetto al bancone presso l'attività di famiglia

Sviluppatore di software
C&B Servizi Informatici [29/11/2016 – 28/06/2017]

Città: Casalmoro (Mantova)
Paese: Italia

Stage esterno all'università svolto durante il periodo di studi: finito il periodo stabilito, possibilità di assunzione non accettata per scelta di proseguire gli studi ed inizio del corso di laurea magistrale in ingegneria informatica. Occasionali collaborazioni a progetto nel 2020/2021.

Attività svolte:
- Attività di sviluppo e debug di software in ambiente Microsoft e Web: Visual Basic .NET, Visual Basic 6, ASP.NET, Javascript, Microsoft SQL Server e Access.
- Attività di installazione e manutenzione reti locali, server dipartimentali e apparati di rete.
- Affiancamento al personale in attività di installazione e manutenzione di personal computer e periferiche sia presso la nostra sede che presso la sede del cliente.

Addetto al bancone e cameriere
Camping Village San Francesco [30/05/2014 – 29/08/2014]

Città: Desenzano del Garda (Brescia)
Paese: Italia

Attività svolte:
- Addetto al bancone e cameriere per la stagione estiva

ISTRUZIONE E FORMAZIONE

Dottorato di Ricerca - Technology for Health
Università degli Studi di Brescia [01/12/2022 – Attuale]

Indirizzo: Via Branze 38, 25123 Brescia (Italia)
Sito web: https://tech4health.unibs.it/

Abilitazione alla professione di ingegnere dell'informazione
Università degli Studi di Brescia [22/07/2020 – 22/07/2020]

Indirizzo: Via Branze 38, 25123 Brescia (Italia)
Sito web: https://www.unibs.it/

Laurea Magistrale, Ingegneria Informatica
Università degli Studi di Brescia [2017 – 2019]

Indirizzo: Via Branze 38, 25123 Brescia (Italia)
Sito web: https://www.unibs.it/
Voto finale: 107/110

Tesi: A domain-independent environment for user-oriented programming of collaborative robots
Link: https://www.sciencedirect.com/science/article/abs/pii/S073658452100106X

Laurea Triennale, Ingegneria Informatica
Università degli Studi di Brescia [2012 – 2016]

Indirizzo: Via Branze 38, 25123 Brescia (Italia)
Sito web: https://www.unibs.it/
Tesi: Verifica di vulnerabilità dei sistemi: concetti generali e analisi dello srumento OpenVAS

Diploma Liceo Scientifico Tecnologico
Istituto Istruzione Superiore Blaise Pascal [09/2007 – 06/2012]

Indirizzo: Via Solferino 92, 25025 Manerbio (Brescia) (Italia)
Sito web: https://www.iis-pascal.edu.it/
Tesi: Concetti di chimica nucleare ed analisi dei disastri di Černobyl' e Fukushima

COMPETENZE LINGUISTICHE

Lingua madre: **Italiano**

Altre lingue:

Inglese
ASCOLTO B2 **LETTURA** B2 **SCRITTURA** B2
PRODUZIONE ORALE B2 **INTERAZIONE ORALE** B2

Francese
ASCOLTO A1 **LETTURA** A1 **SCRITTURA** A1
PRODUZIONE ORALE A1 **INTERAZIONE ORALE** A1

Livelli: A1 e A2: Livello elementare B1 e B2: Livello intermedio C1 e C2: Livello avanzato

COMPETENZE DIGITALI

Front-End React Redux JavaScript Bootstrap HTML5 CSS3 / Utilizzo base di git (git bash, gitlab, ...) / Node.Js / Appassionato di robotica, intelligenza artificiale e machine learning / Interfaccia uomo-macchina / Ricerca e Sviluppo

PUBBLICAZIONI

A hybrid approach to user-oriented programming of collaborative robots
[2022]

Robotics and Computer-Integrated Manufacturing, Volume 73, February 2022

The research reported in this paper proposes a new approach to collaborative robots that aims at improving the simplicity and efficiency of the programming task for non-technical users. It is grounded on three standpoints: (i) an elementary and disciplined paradigm for robot programming, called the simple programming journey, (ii) a hybrid interaction mode where robot tasks can be programmed using a natural language chat and, if necessary, can be completed and finalized through a block-based interface, and (iii) a robust cognitive match between the mental models of the user and the programming interface. The proposed approach has been implemented and tested through the development of a prototype programming environment called CAPIRCI, which can be tailored to different application domains through the definition of objects, locations, and actions. CAPIRCI has been tested by real users with a COBOTTA robot by DENSO WAVE Ltd. Two experimental tests have been carried out in order to validate the novel approach proposed and to assess its impact on end-user programming. The results obtained show that a hybrid approach exploiting both natural language dialogue and block-based interaction can help make the programming task easy and efficient for non-technical users.

Human-robot collaboration in healthcare: new programming and interaction techniques
[2023]

IS-EUD 2023, 9th International Symposium on End-User Development, 06-08 June 2023, Cagliari, Italy

In the field of healthcare, collaborative robots can be utilized to enhance productivity and efficiency. This research will delve into the development and investigation of new techniques to enable effective interaction with

collaborative robots. The focus will be on designing innovative techniques that can be easily understood and implemented by individuals with no technical knowledge in computer science or robotics. Collaborative robots can be used to automate repetitive tasks, allowing healthcare professionals to focus on more complex procedures. For instance, by leveraging this technology, it is possible to significantly increase the efficiency and speed of therapy preparation, ultimately improving patient outcomes. The study will explore various approaches to simplify and streamline the programming process, reducing the need for technical knowledge and expertise.

Can ChatGPT Support End-User Development of Robot Programs?

[2023]

CoPDA 2023, Cultures of Partecipation in the Digital Age, International workshop @ IS-EUD 2023

In this paper, we investigate the use of OpenAI ChatGPT to improve the natural language understanding of an End-User Development environment, called CAPIRCI, supporting users neither expert in computer programming nor expert in robotics to create programs for a collaborative robot. The integration of ChatGPT in CAPIRCI is studied to be transparent for the user, who will be allowed to check and correct that program description generated by ChatGPT, by interacting with an intuitive block-based interface, according to a Human-Centered Artificial Intelligence design approach.

Designing Human-Robot Collaboration for the Preparation of Personalized Medicines

[2023]

GoodIT 2023, International Conference on Information Technology for Social Good, Lisbon, Portugal

Advancements in robotics and automation technologies have the potential to enable breakthrough innovations in a variety of industries, and the pharmaceutical sector is no exception. The preparation of galenic formulations, involving the compounding and dispensing of medications, when personalized medicines are needed, e.g., to overcome allergy problems, is a critical process in the field of small scale pharmaceutical manufacturing. Traditionally, this process has relied solely on human expertise of pharmacists and their manual labor, which can be time-consuming, prone to errors, and subject to variations in quality. To overcome these limitations, the use of collaborative robots is envisaged in our project. A collaborative robot can in fact work with the pharmacist synergistically, by improving accuracy and increasing productivity. However, the main challenge is providing the pharmacists with an interactive system that supports them in robot programming. In this paper, we analyze the problem from the users' point of view and propose preliminary low-fidelity prototypes of an interactive system suitable to pharmacists' needs and skills.

A Systematic Review on Pill and Medication Dispensers from a Human-Centered Perspective

[2023]

Journal of Healthcare Informatics Research, Springer - Peer Review Stage

As medication adherence represents a critical challenge in healthcare, pill and medication dispensers have gained increasing attention as potential solutions to promote adherence and improve patient outcomes. Following the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) methodology, we carried out a systematic literature review on papers indexed in Scopus, which present solutions for pill or medication dispensers. Given the importance of user acceptance for these solutions, the research questions of the survey are driven by a human-centered perspective. We first provide an overview of the different solutions, classifying them according to their stage of development. We then analyze each solution considering its hardware/software architecture. Finally, we review the characteristics of user interfaces designed for interacting with pill and medication dispensers and analyze the involvement of different types of users in dispenser management. On the basis of this analysis, we draw findings and indications for future research that are aimed to provide insights to healthcare professionals, researchers, and designers who are interested in developing and using pill and medication dispensers.

PROGETTI

Gestionale per carpenteria metallica

[12/2020 – 08/2022]

Gestionale per carpenteria metallica sviluppato con tecnologie web:

• Front-end: Typescript, Electron, React, Redux, Webpack, Ant Design, Styled-Components

• Back-end: Typescript, Node.js, Express, Sequelize, Morgan, JWT, bcrypt

• Database: MariaDB

• Controllo di versione con Git (GitHub)

• Supporto (HW/SW): Raspberry Pi, GitHub Releases, Husky, PM2

CAPIRCI (Chat And Program Industrial Robots through Convenient Interaction)

[02/2019 – 10/2019]

CAPIRCI (Chat And Program Industrial Robots through Convenient Interaction), un'applicazione web multimodale che supporta gli utenti finali, senza alcuna competenza in informatica, per definire e modificare le attività da eseguire da robot collaborativi. L'applicazione fornisce due modalità di interazione, la prima basata su un'interfaccia di chat, la seconda che presenta un linguaggio di programmazione visuale ispirato a soluzioni basate su blocchi, ma adattato al dominio specifico.

Tecnologie utilizzate: Python, Django, OpenCV, Javascript, Ajax, Css, Standford Parser, ORiN e B-cap.

Robot utilizzato: Denso Cobotta con camera Canon N10-W02 e Denso VS.

PallavoloRemedello.it

[08/2018 – 05/2019]

Sito Web per la squadra di pallavolo A.S.D. Pallavolo Remedello.

Tecnologie utilizzate: Javascript, jQuery, Ajax, PHP e CSS.

ONORIFICENZE E RICONOSCIMENTI

MongoDB Node.js Developer Path

MongoDB [02/2023]
Link: https://learn.mongodb.com/learn/learning-path/mongodb-nodejs-developer-path

Postman API Fundamentals Student Expert

Postman [01/2023]

ID credenziale: WIoUhlzfQOqVq57BMYKK_Q

Link: https://badgr.com/public/assertions/WIoUhlzfQOqVq57BMYKK_Q?
identity__email=l.gargioni@studenti.unibs.it

Machine Learning

MIT Open Learning [08/2022]
Link: https://openlearninglibrary.mit.edu/courses/course-v1:MITx+6.036+1T2019/course/

How to Write and Publish a Scientific Paper (Project-Centered Course)

École Polytechnique [04/2022]

ID credenziale: 46CKHJA5BZRE

Link: https://www.coursera.org/account/accomplishments/verify/46CKHJA5BZRE

Intelligenza Artificiale e sostenibilità

Fastweb Digital Academy [04/2022]

ID credenziale: 63382067693d40f6c9f071c284da76a972873f62

Link: https://learn.fastwebdigital.academy/badges/badge.php?
hash=63382067693d40f6c9f071c284da76a972873f62

Elements of AI

University of Helsinki [11/2021]

ID credenziale: 7kas3px5o4q

Link: https://certificates.mooc.fi/validate/7kas3px5o4q

Machine Learning

Stanford University [09/2020]

ID credenziale: G5J6SZ2UXRDX

Link: https://www.coursera.org/account/accomplishments/verify/G5J6SZ2UXRDX

Basic Certification

3CX [09/2020]

ID credenziale: ezftPKvV8p

Link: https://customer.3cx.com/documents/CertificationSharing.aspx?c=ezftPKvV8p

Cisco Networking Academy

Cisco Systems [04/2020]

- Intro to IoE
- Introduction to Cybersecurity
- Cybersecurity Essentials
- NDG Linux Essentials
- NDG Linux Unhatched

Link: https://www.youracclaim.com/badges/a4cdaf40-7ae7-4422-a89b-5b9ca09f7836

ISE I Integrated Skills in English - B1 of the CEFR

Trinity College London [03/2012]

European Computer Driving License (ECDL)

AICA - Associazione Italiana per l'Informatica e il Calcolo Automatico [06/2011]

HOBBY E INTERESSI

Pallavolo

Calcio

Softair

Lettura

RETI E AFFILIAZIONI

Associazione Alumni Università degli Studi di Brescia

[Brescia, 11/2022 – Attuale]
Link: https://alumni.unibs.it/documents/badges/badge.php?
hash=02608da02a1dbd8ef89a8bd5b40403df2a766f75